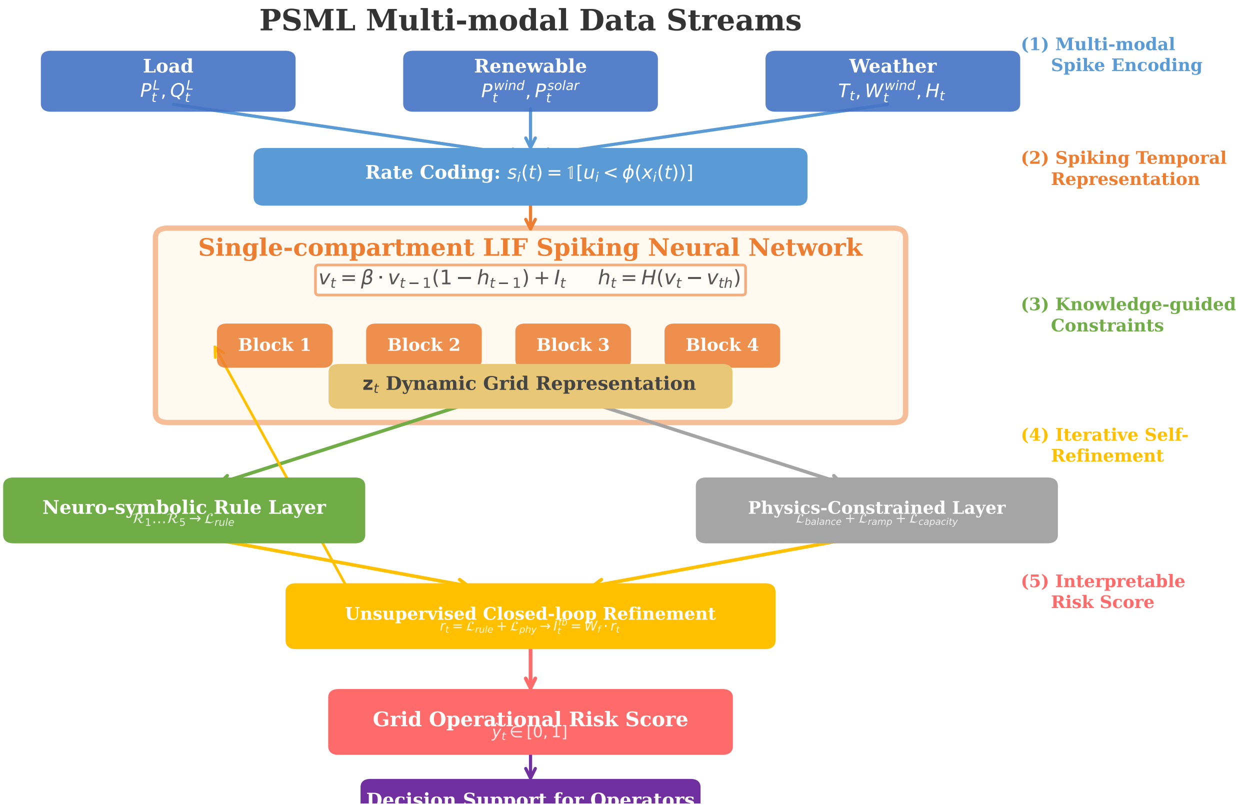
PSML Multi-modal Data Streams
Load
P_t^L, Q_t^L
Renewable
P_t^{wind}, P_t^{solar}
Weather
T_t, W_t^{wind}, H_t
(1) Multi-modal Spike Encoding
Rate Coding: $s_i(t) = \mathbb{1}[u_i < \phi(x_i(t))]$
(2) Spiking Temporal Representation
Single-compartment LIF Spiking Neural Network
$v_t = \beta \cdot v_{t-1}(1 - h_{t-1}) + I_t \quad h_t = H(v_t - v_{th})$
(3) Knowledge-guided Constraints
Block 1
Block 2
Block 3
Block 4
$\mathbf{z}_t$ Dynamic Grid Representation
(4) Iterative Self-Refinement
Neuro-symbolic Rule Layer
$\mathcal{R}_1 \ldots \mathcal{R}_5 \to \mathcal{L}_{rule}$
Physics-Constrained Layer
$\mathcal{L}_{balance} + \mathcal{L}_{ramp} + \mathcal{L}_{capacity}$
(5) Interpretable Risk Score
Unsupervised Closed-loop Refinement
$r_t = \mathcal{L}_{rule} + \mathcal{L}_{phy} \to I_t^{fb} = W_f \cdot r_t$
Grid Operational Risk Score
$y_t \in [0, 1]$
Decision Support for Operators